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Report

RITM-200 Dynamics and Control

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Abstract

This work analyzes the dynamic behavior of the RITM-200 SMR-type nuclear reactor following operational perturbations, through the development of a coupled numerical model implemented in Python.

The model integrates three physical subsystems: point-kinetics neutronics with six groups of delayed neutrons, core thermal-hydraulics with three radial fuel nodes, and a one-dimensional finite-volume model for the counter-current concentric-tube heat exchanger. The dynamics of the secondary-circuit pressure are described through a first-order phenomenological model, derived from the internal energy balance of the secondary fluid.

Time integration of the coupled system is carried out with an unconditionally stable implicit Euler scheme for the heat exchanger, and with the **LSODA** method for the core's differential equations, suited to stiff systems. The two models are coupled through the *hot leg* and *cold leg* temperatures of the primary circuit, updated at every time step.

Two transient scenarios are analyzed: an increase in the secondary-circuit mass flow rate, illustrating the *load following* capability characteristic of PWR reactors, and a control-rod insertion into the core, demonstrating upstream power control. In both cases, the results show correct propagation of the effects between subsystems and stability of the new equilibrium state reached, confirming the validity of the model developed for the analysis of operational transients in SMR reactors.

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Chapter 1

Introduction

1.1 Reactor description

The RITM-200 is a Generation III+ SMR-type reactor designed to produce up to 55MWe according to PWR technology. It has a design lifetime of 60 years; the fuel used is uranium enriched to 14.06%, and its thermodynamic cycle depends on the reactor's application (Arctic naval propulsion, electricity generation on floating power plants, or future modular land-based plants), requiring refueling every 6 years for land-based applications or 10 years for naval propulsion. Having originally been designed for icebreaker ships, it features a compact and lightweight structure: all major components are in fact incorporated within the vessel — the core is located in the lower central part, the 4 heat exchangers are arranged radially around the core, and pressure is regulated in the upper part of the vessel without the need for a pressurizer. This configuration differs from that typical of large PWR plants; the reduction in piping offers an advantage not only in economic terms but also from a safety standpoint.

Table 1.1: Operating values

Primary operating pressure	15.7 MPa
Coolant inlet temperature	277 °C
Coolant outlet temperature	313 °C
Coolant mass flow rate	3250 t/h
Rated thermal power	175 MW
Electric power	2×55 MW
Secondary pressure	3.4 MPa

1.2 Numerical methods

The problem was solved using Python, which facilitates the solution of stiff systems of ordinary differential equations (i.e., systems in which two variables evolve on very different time scales), nonlinear systems for steady-state conditions, and the spatial

discretization of the transport equations in the heat exchanger. The code employs the `numPy` and `sciPy` libraries, together with the `pyXSteam` library for computing the water properties.

1.2.1 Integration of the core ODEs

The core equations combine six-delayed-neutron-group kinetics with a thermal-hydraulic model using three radial nodes for the fuel, the cladding, and the coolant, for a total of $6 + 5 = 11$ coupled differential equations. The system exhibits a high degree of *stiffness*, due to the coexistence of very different time constants: the prompt-neutron lifetime $\Lambda = 10^{-4}$ s versus the thermal time constants of the fuel and coolant, of the order of tens of seconds. To cope with this stiffness, the `solve_ivp` function from `scipy.integrate` is used with the LSODA method (Livermore Solver for ODEs with Automatic stiffness detection and method switching), which dynamically switches between an Adams-Moulton integrator (suited to non-stiff problems) and a BDF method (Backward Differentiation Formula, implicit, first-to-fifth order) depending on the local condition number of the Jacobian. The integration tolerances are set to `rtol` = 10^{-6} and `atol` = 10^{-8} , ensuring adequate numerical accuracy without excessive computational cost.

1.2.2 Coupled steady-state solution

The coupled core-steam-generator steady state is sought through two levels of iteration. At the inner level, the nonlinear system of five algebraic equations for the fuel temperatures (T_1, T_2, T_3), the cladding (T_d), and the coolant (T_p) is solved with the `fsolve` function from `scipy.optimize`, which implements a Powell-type hybrid method based on finite differences for the approximation of the Jacobian. At the outer level, the cold-leg temperature that simultaneously balances the core and the exchanger is sought with `root_scalar` (again from `scipy.optimize`), applying the Brent-Dekker method (`method='brentq'`), which guarantees convergence in a finite number of steps over an initial interval defined between $T_{s,in}$ and T_{hot} .

1.2.3 Implicit scheme for the steam generator

The steam generator is modeled as a counter-current coaxial-tube exchanger, discretized axially into $N = 30$ control volumes. Time integration of the energy-balance equations for the primary and secondary sides is performed with an *implicit Euler* scheme on each finite cell, which, being unconditionally stable, is not subject to *CFL*-type constraints. The *upwind* structure of the counter-current transport allows the problem to be factorized into two sequential *sweeps* with no matrix inversion: at each step, the updated temperature of the downstream cell is already available, reducing the system to a simple cell-by-cell forward substitution.

1.2.4 Warmup procedure

Initially, an implicit *warmup* phase of 1000 steps with $\Delta t_{wu} = 1$ s is carried out, aimed at bringing the discrete exchanger profiles to the steady state consistent with

the numerical scheme adopted. At each warmup step, a predict-and-correct cycle is performed on the *SG*–core coupling: given an estimate of the *cold leg* temperature from the current primary profile, the corrector updates it using the just-integrated *SG* profile, and the core is integrated with `solve_ivp`/LSODA over the interval $[0, \Delta t_{wu}]$. Since the implicit scheme is not subject to *CFL* stability constraints, the step Δt_{wu} can be chosen to be $\sim 10\times$ the primary-coolant transit time through the steam generator, reducing the number of steps needed for convergence by more than two orders of magnitude compared with an explicit scheme.

Chapter 2

Neutronics

In this chapter we analyze the reactor’s neutron kinetics. In this context, neutronic analysis makes it possible to simulate the time response of the neutron density, on which the evolution of the thermal power generated in the core directly depends. Since the reactor under study, being an SMR, is designed to operate with high standards of stability and passive safety, it is necessary to study how the system responds dynamically to sudden external or operational perturbations. The model focuses on computing the total reactivity of the system (ρ) and on solving the point-kinetics differential equations. Reactivity is expressed by evaluating the feedback coefficients related to variations in fuel temperature (α_f), moderator temperature (α_m), and control-rod insertion (α_h). Three-dimensional spatial modeling of the neutron flux was, however, neglected, since the zero-dimensional (0D) approach, combined with the use of six groups of delayed-neutron precursors, captures the overall transient behavior with sufficient accuracy for this analysis.

2.1 Physical phenomenology and feedback

The dynamic behavior of a nuclear reactor, particularly in the case of compact systems such as the RITM-200, is governed by a close nonlinear interaction between the neutron population and the thermal-hydraulic quantities that characterize the core. The guiding principle governing the overall stability of the system is the concept of reactivity feedback. Reactivity is a measure of the reactor’s deviation from the steady critical condition: thermal, geometric, or mechanical variations within the core instantaneously alter the balance between neutrons produced by fission and those lost to capture or leakage, consequently changing the neutron density and the thermal power generated.

2.1.1 Feedback mechanisms

The mechanisms governing this balance are divided into intrinsic thermophysical contributions of the reactor, and exogenous ones, i.e., control-related contributions:

Intrinsic feedback mechanisms

Fuel Doppler effect

This constitutes the main passive safety mechanism: a rise in fuel temperature increases the thermal agitation of the fuel nuclei, broadening their energy absorption windows (widening the resonance spectrum) due to the relative motion with respect to the incident neutrons. This broadening increases parasitic neutron capture in the core, generating an immediate insertion of negative reactivity that self-dampens and instantaneously stabilizes the reactor in the event of power spikes.

Moderator feedback

A rise in water temperature causes it to expand and its density to fall accordingly. Lower density degrades the fluid’s moderating capacity: neutrons retain higher energies, increasing the probability of leakage or of capture before thermalization. This introduces negative reactivity. Conversely, a sudden cooling (which increases the water density) introduces positive reactivity.

Exogenous feedback mechanisms (Control systems)

Control rods The insertion of absorber materials into the core locally increases capture losses. In lumped-parameter models, this geometric effect is described through a global reactivity term that depends on the rod insertion height.

2.1.2 Delayed neutrons and precursors

Although most neutrons are emitted almost instantaneously at the moment of fission, a small fraction β (about 0.65%) is released with a delay, from the radioactive decay of specific fission fragments called precursors. Since these isotopes decay with characteristic times ranging from fractions of a second to nearly a minute, this phenomenology is modeled by grouping them into six main time-based families. Although the fraction β is numerically small, delayed neutrons are the linchpin of plant controllability: they extend the reactor’s dynamic inertia, stretching the power-exursion timescales from fractions of a millisecond to tens of seconds, and thereby making intervention by the mechanical control systems materially possible.

2.2 Mathematical model

The reactor’s neutronic behavior was modeled through the point-kinetics equations with delayed neutrons, assuming that the spatial distribution of the neutron flux remains unchanged and that the time evolution can be described solely through the variation of the neutron density. In this approach, the system dynamics are governed by a differential equation for the neutron density $n(t)$ and by a set of equations for the concentrations $C_i(t)$ of the delayed-neutron precursors.

$$\frac{dn(t)}{dt} = \frac{\rho - \beta}{\Lambda} n(t) + \sum_{i=1}^6 \lambda_i C_i(t) \quad (2.1)$$

The evolution of the precursor concentrations belonging to the six delayed-neutron groups is instead described by:

$$\frac{dC_i(t)}{dt} = \frac{\beta_i}{\Lambda} n(t) - \lambda_i C_i(t) \quad i = 1, \dots, 6 \quad (2.2)$$

where ρ represents the system reactivity, β the total delayed-neutron fraction, β_i the fraction associated with the i -th group, λ_i the decay constant of the corresponding precursors, and Λ the mean prompt-neutron lifetime. These parameters are introduced into the code as numerical constants based on typical values for thermal reactors.

Reactivity was modeled by considering three feedback contributions: fuel temperature, moderator temperature, and control-rod position. It is expressed as a linear combination of the deviations from the reference conditions, using the following equation:

$$\rho = \alpha_f(T_f - T_{f0}) + \alpha_m(T_c - T_{c0}) + \alpha_h(h - h_0) \quad (2.3)$$

where α_f , α_m , α_h are the reactivity coefficients associated, respectively, with the fuel temperature, the moderator temperature, and the control-rod height. These coefficients are assumed constant over the thermal range explored, introducing a first-order approximation. This linearization is acceptable for operational transients that do not deviate drastically from nominal conditions, and it makes it possible to describe the dynamic response of the system accurately while maintaining high computational efficiency. In the implemented dynamic model, the total reactivity ρ is expressed through the decomposition between the intrinsic thermal-feedback contributions of the core and the exogenous action of the control systems. More specifically, the thermal variations of the fuel and moderator constitute the system's stable intrinsic feedbacks, while the rod displacement represents the exogenous control input.

In this treatment, the reference temperatures considered are equal to the reactor's design temperatures, namely a mean fuel temperature of 550 °C and a moderator temperature of 310 °C. These values were later modified in Section 3.2, in which the steady-state values are computed through the reactor's thermal-hydraulic equations.

The initial concentrations of the precursor families are calculated by imposing the steady state in equation (2.2)

$$C_i(0) = \frac{\beta_i}{\lambda_i \Lambda} n(0) \quad (2.4)$$

Below is a brief legend of the data conventions used:

- $n(t)$: neutron density
- $C_i(t)$: concentration of group- i precursors
- ρ : reactivity
- β : total delayed-neutron fraction
- β_i : fraction of delayed-neutron group i
- λ_i : precursor decay constant

- Λ : mean prompt-neutron lifetime
- T_f, T_c : fuel and moderator temperature
- h : control-rod position

2.3 Results

In order to validate the model and observe the core dynamics, the system of differential equations was integrated numerically starting from the steady-state equilibrium conditions ($T_{f0} = 550$ °C, $T_{c0} = 310$ °C, $n_0 = 1.0$). To test the stability of the system, step perturbations were imposed on the thermal feedback parameters (T_f, T_c) and on the mechanical parameter (h), analyzing the resulting time evolution of $n(t)$.

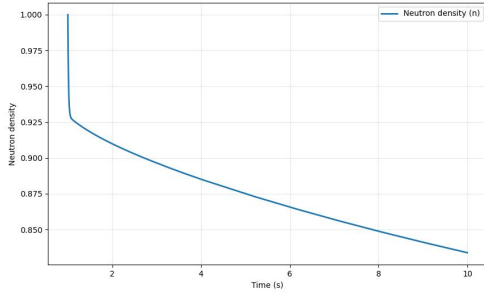


Figure 2.1: $T_f = 600^\circ\text{C}, T_c = 305^\circ\text{C}, h = 0\text{cm}$

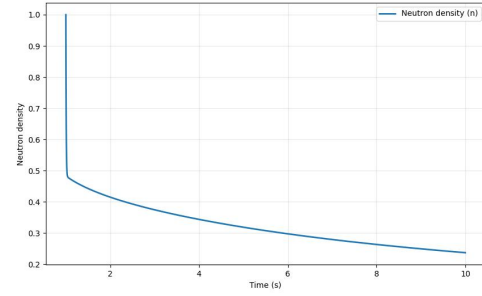


Figure 2.2: $T_f = 550^\circ\text{C}, T_c = 310^\circ\text{C}, h = 1\text{cm}$

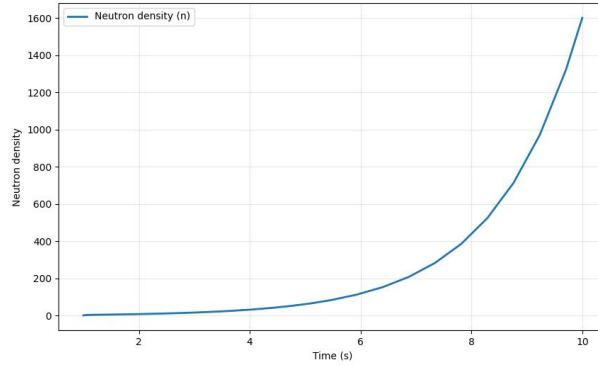


Figure 2.3: $T_f = 660^\circ\text{C}, T_c = 240^\circ\text{C}, h = 0$

In the first figure 2.1, a 50 °C increase in fuel temperature together with a 5 °C decrease in moderator temperature leads to a decrease in neutron density: the Doppler effect has a greater impact than the increase in moderator density.

In the second figure 2.2, a lowering of the control rods results in a natural drop in neutron density, driven by the exogenous intervention on the chain reaction.

In the last plot [2.3](#), a 110°C increase in fuel temperature together with a 70°C decrease in moderator temperature leads to an exponential growth of the neutron density: this occurs because the neutronic contribution given by the densification of the moderator far exceeds the reaction damping caused by the Doppler effect.

Having now defined and validated the purely neutronic response of the reactor to external perturbations, the next step will be to integrate these equations within the broader thermal-hydraulic framework. In the following chapters, the two models will therefore be coupled, in order to capture the true interdependent nature of the reactor's operational transients.

Chapter 3

Thermal-hydraulics

The objective of the developed model is to build a simplified, yet physically consistent, thermal-hydraulic representation of the reactor. The analysis aims to study the system's transient behavior following a power variation, introduced through a positive step of 10%, and to evaluate the dynamic response of the fuel and of the coolant fluid. A lumped-parameter approach was adopted, in which the system's behavior is described through the average values of the variables of interest, considered spatially uniform and time-dependent.

3.1 Model definition

To account for the dependence of the conductive heat-transfer coefficient and of the fuel's specific heat, the core was divided into 3 concentric volumes, where the inner volume is a solid cylinder and the two outer ones are annular cylinders, each with its own temperature and thermal conductivity (temperature-dependent). For the subdivision of the volumes, the chosen radii were computed starting from the fuel-rod radius r_f . In the central zone $r < r_1$, in the intermediate zone $r_1 < r < r_2$, and in the outer zone $r_2 < r < r_f$, with:

$$r_1 = \frac{r_f}{\sqrt{3}}, \quad r_2 = r_f \sqrt{\frac{2}{3}} \quad (3.1)$$

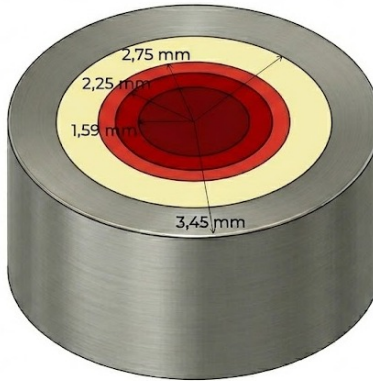


Figure 3.1: Subdivision of the fuel element

The reason for this subdivision lies in the large temperature variation within the fuel. Although the diameter is on the order of a few millimeters, the variation between the center and the surface of the fuel is about 300 °C.

The total thermal resistance was modeled as a series sum of the conduction and convection contributions:

$$R_{tot} = R_{fuel} + R_{gap} + R_{clad} + R_{conv} \quad (3.2)$$

with:

$$R_{fuel} = \frac{1}{4\pi H k_f(T_f)}, \quad (3.3)$$

$$R_{gap} = \frac{1}{2\pi r_f H h_{gap}}, \quad (3.4)$$

$$R_{clad} = \frac{\ln\left(\frac{r_{ce}}{r_{ci}}\right)}{2\pi H k_{clad}}, \quad (3.5)$$

$$R_{conv} = \frac{1}{2\pi r_{ce} H h_c}. \quad (3.6)$$

The resistances of the helium gap and of the *cladding* are fixed: the former with a convective heat-transfer coefficient set to $1593.77 \frac{W}{m^2 \cdot K}$, and the latter with a conductivity of $17 \frac{W}{m \cdot K}$. Finally, the water's convective resistance depends on the reactor geometry and on the water properties. In this latter case, the contribution of the term

$$2 \cdot \dot{m}_{channel} \cdot c_{p,c}(T_c - T_{inlet}) \quad (3.7)$$

was also taken into account. The convective heat-transfer coefficient h_c was evaluated using the Dittus-Boelter correlation, corrected for rod bundles, as a function of the dimensionless Reynolds and Prandtl numbers:

$$Nu = 0.023 Re^{0.8} Pr^{0.4} C_{weisman} \quad (3.8)$$

$$h_c = \frac{Nu k}{D_h} \quad (3.9)$$

$$Re = \frac{\rho v D_h}{\mu} \quad (3.10)$$

$$Pr = \frac{\mu c_p}{k} \quad (3.11)$$

where the hydraulic diameter D_h

$$D_h = D \left(\frac{2\sqrt{3}}{\pi} \left(\frac{P}{D} \right)^2 - 1 \right) \quad (3.12)$$

was derived from the geometry of the triangular lattice.

The fuel's thermal capacity was modeled as a function of temperature using the semi-empirical Fink-Touloukian correlation:

$$c_f(T) = 4.186 \cdot 10^{-2} (710 + 0.6 \cdot T + 4 \cdot \left(\frac{T}{1000} \right)^7) \quad (3.13)$$

The thermal conductivity coefficient k is likewise expressed as a function of temperature:

$$k = \frac{100}{11.8 + 0.0238 \cdot T} + 8.775 \cdot 10^{-15} T^3 \quad (3.14)$$

The model focuses on the thermal dynamics of the fuel, described through a non-steady energy balance. In particular, the time variation of the fuel temperature is given by the difference between the power generated by fission and the heat transferred to the coolant fluid.

3.2 Initial conditions

Before examining the evolution of the quantities of interest, we determined the actual steady-state temperatures for our model, in order to obtain greater accuracy in the initial values of the transient.

The temperature values under steady-state conditions were determined by means of energy balances. Given the total thermal power generated by the *fuel*, equal to 175MW, and distributed among the three zones through a weighted average over the respective volumes (3.15), this power must equal the exchanged power, expressed as the ratio between the temperature difference and the corresponding thermal resistance.

In steady-state conditions the time derivatives vanish, and the ODE system reduces to a nonlinear algebraic system: for each node, the power generated or entering (or both) must equal the power leaving. The power generated in the i -th fuel volume is given by:

$$\dot{Q}_i = P_{\text{rod}} \frac{V_i}{V_1 + V_2 + V_3}, \quad i = 1, 2, 3 \quad (3.15)$$

For the central volume:

$$\dot{Q}_1 = \dot{Q}_{12} \implies P_{\text{rod}} \frac{V_1}{V_1 + V_2 + V_3} = \frac{T_1 - T_2}{R_{12}(T_1)} \quad (3.16)$$

Balances for the intermediate volume, the outer volume, the cladding, and the coolant are found analogously, where the outlet term is the flow rate computed as: $2 \dot{m}_{ch} c_{p,c}(T_c - T_{in})$. The resulting system is solved numerically, using as a starting point an accurate initial estimate obtained from the lumped total-resistance model.

3.3 Transient

For the fuel, the energy balance is expressed as:

$$m_f c_{pf}(T_f) \frac{dT_f}{dt} = P(t) - \frac{T_f - T_c}{R_{tot}(T_f)} \quad (3.17)$$

where m_f is the fuel mass, $c_{pf}(T_f)$ the temperature-dependent specific heat, $P(t)$ the thermal power generated per rod, and R_{tot} the equivalent thermal resistance between

fuel and coolant.

For the coolant, on the other hand, the energy balance takes the form:

$$m_c c_{pc} \frac{dT_c}{dt} = \frac{T_f - T_c}{R_{tot}} - 2\dot{m}c_p(T_c - T_{in}) \quad (3.18)$$

where m_c is the mass of fluid in the channel, c_{pc} the specific heat, $\dot{m}$ the mass flow rate per channel, and T_{in} the coolant inlet temperature.

The simulated transient consists of a 10% step increase in power at time $t = 10$ s:

$$P(t) = \begin{cases} P_0 & t \leq 10 \text{ s} \\ 1.1 \cdot P_0 & t > 10 \text{ s} \end{cases} \quad (3.19)$$

The adopted approach makes it possible to capture the coupled fuel–coolant thermal dynamics, and is particularly well suited to fast transient analyses and parametric studies.

The following figures show the results, allowing the evolution of the fuel and coolant temperature over time to be observed, and the speed and intensity of the system's thermal response to be evaluated.

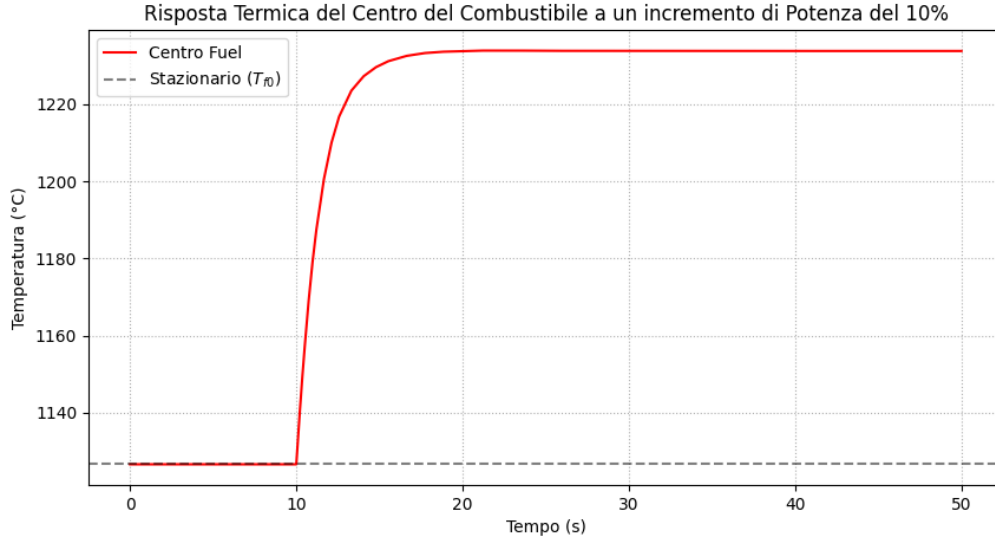


Figure 3.2: Evolution of the fuel centerline temperature (T_f) following a 10% increase in thermal power

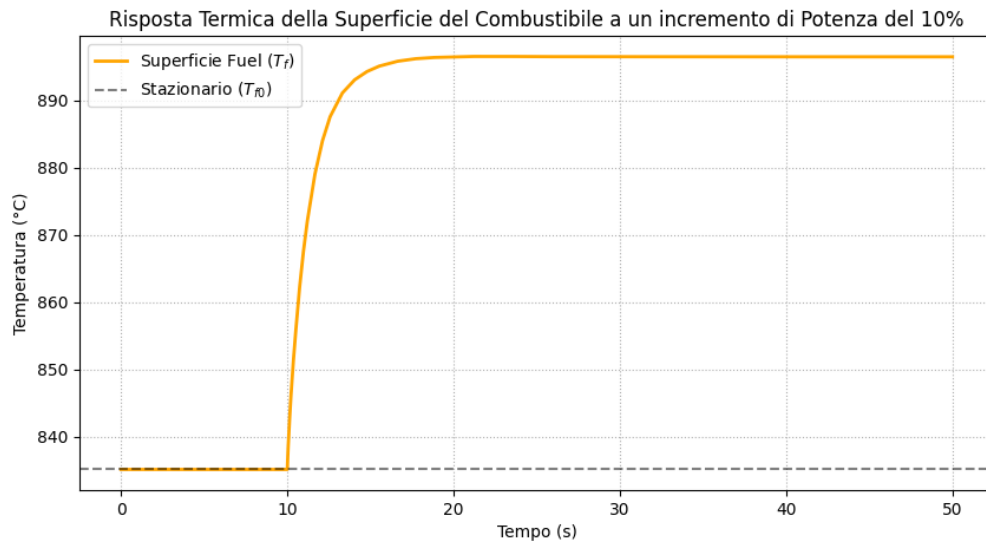


Figure 3.3: Evolution of the fuel surface temperature following a 10% increase in thermal power

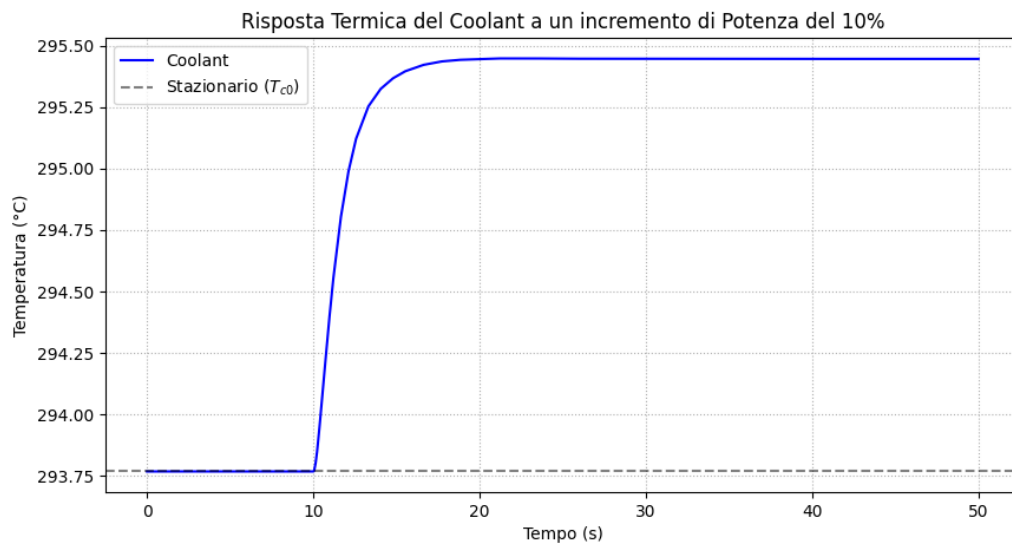


Figure 3.4: Evolution of the coolant temperature following a 10% increase in thermal power

3.4 Neutronics and thermal-hydraulics

The final phase of the work concerns the coupling between neutronics and thermal-hydraulics, aimed at analyzing how the effects of a transient originating in one of the two subsystems propagate to the characteristic quantities of the other.

In the previous sections the two aspects were treated separately: the neutronic response to a change in control-rod position, and the thermal response of the fuel to a step increase in generated power. In this section, the two transients are instead simulated in a coupled way: the control-rod position, assigned by the user as the initial condition at $t = 0$ s, introduces a reactivity perturbation that changes the neutronic power; at $t = 30$ s, a 10% step increase in the primary mass flow rate is superimposed:

$$\dot{m}(t) = \begin{cases} \dot{m}_0 & t < 30 \text{ s} \\ 1.1 \dot{m}_0 & t \geq 30 \text{ s} \end{cases} \quad (3.20)$$

This second transient cools the primary coolant, lowering T_p and introducing positive reactivity through the coefficient α_m , which tends to increase the neutronic power. The coupling is achieved by introducing into the reactivity the dependence on the temperatures computed by the thermal model:

$$\rho(t) = \alpha_f(T_2(t) - T_{2,0}) + \alpha_m(T_p(t) - T_{p,0}) + \alpha_h(h - h_0) \quad (3.21)$$

where T_2 is the temperature of the intermediate fuel node, T_p is the mean coolant temperature, and h is the control-rod position assigned at $t = 0$ s. Substituting (3.21) into the point-kinetics equations (2.1)–(2.2), and coupling these to the fuel thermal equations through the source term:

$$q_i(t) = \frac{P_0 n(t) V_i}{V_1 + V_2 + V_3}, \quad i = 1, 2, 3 \quad (3.22)$$

a system of 12 coupled ordinary differential equations is obtained: six for the precursors, one for the neutron density, and five for the temperatures. The integrated response of the system is reported in the following plots:

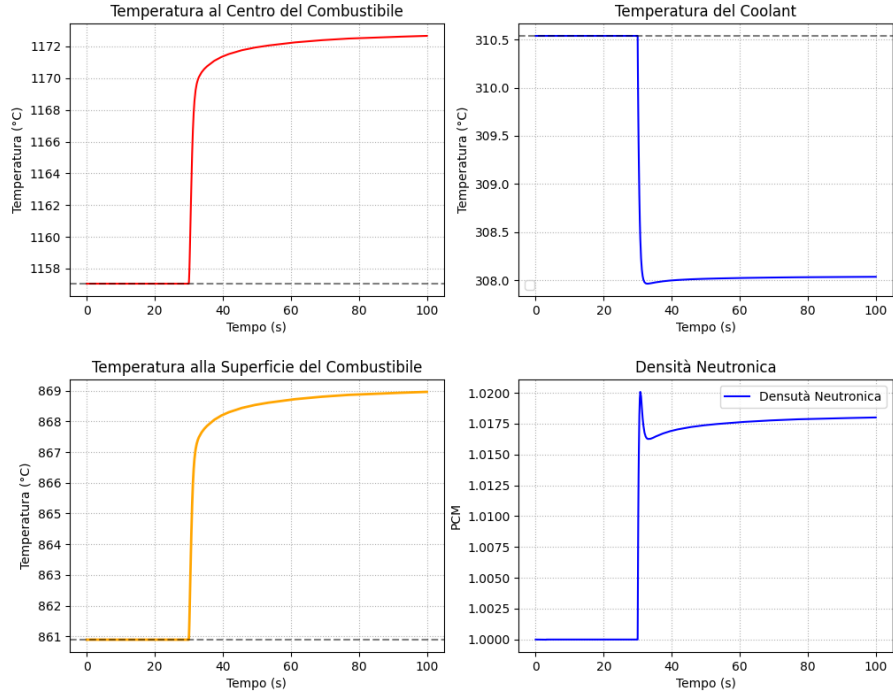


Figure 3.5: Flow-rate increase: 10%, control-rod insertion: $h = 0\text{ cm}$

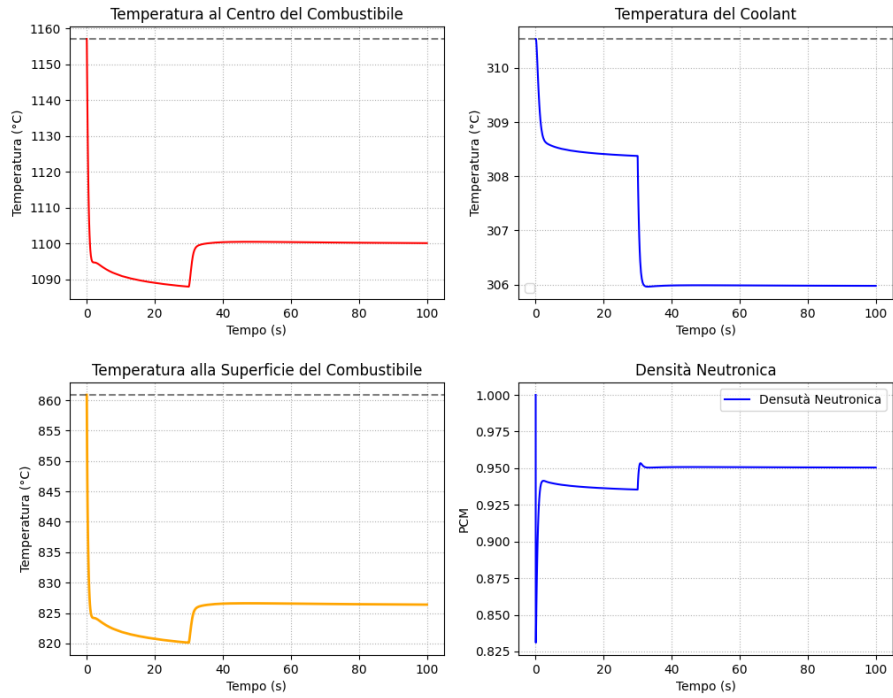


Figure 3.6: Flow-rate increase: 10%, control-rod insertion: $h = 2\text{ cm}$

Chapter 4

Numerical Model of the Cassette-Type Steam Generator

4.1 Introduction

After modeling the core's thermal-hydraulics, this section analyzes the behavior of the heat exchanger. It transfers the thermal power generated by the fission reactions from the primary fluid to the secondary fluid. Since our reactor, being an SMR, does not produce a large amount of heat, the exchanger must be very efficient in order to optimize the usable electric power. The model computes the temperature and enthalpy profiles of the primary and secondary fluids along the axial direction of the exchanger, both under steady-state conditions and after an operational transient induced by a user-defined variation of the secondary flow rate.

The objective of the simulation, in particular, is twofold:

- to determine the steady state of the system (primary temperature field, and secondary enthalpy and temperature) starting from the inlet pressure and temperature/enthalpy conditions of the two fluids;
- to evaluate the dynamic response of the system to a change in the turbine-regulated secondary mass flow rate, quantifying the new equilibrium condition and showing the transient behavior of both the primary outlet temperature and the exchanged power over time.

In our treatment there are 4 once-through, concentric-tube heat exchangers. The modeling of the heat exchanger's resistances is analogous to that adopted for the core resistances; however, the initial part of the exchanger was neglected, since the amount of thermal power exchanged in that section is negligible compared with that exchanged in the rest of the system, as will be shown later.

Table 4.1: Heat exchanger geometric values

Quantity	Cassettes	Sections
Cross-section of exchange tubes	$13 \times 1.5mm$	$8 \times 1.5mm$
Number of cassettes per section	3	-
Number of modules	7	21
Number of heat-exchange tubes in the modules per section	118	
Number of heat-exchange tubes	826	2478
Tube material	PT-7M alloy	
Heat-exchange surface area	$93.4 m^2$	$280 m^2$
Thermal power	14.5 MW	43.7 MW

4.2 Problem geometry and symmetry assumptions

The exchanger consists of three concentric channels, each with cylindrical symmetry about the tube's longitudinal axis z (Figure 4.1):

- the **inner tube**, through which part of the primary fluid flows;
- the **annulus**, the circular ring between the inner tube and the outer tube, through which the secondary fluid flows;
- the **shell** (*mantello*), outside the outer tube, through which the remaining part of the primary fluid flows, with the tubes arranged in a triangular pitch.

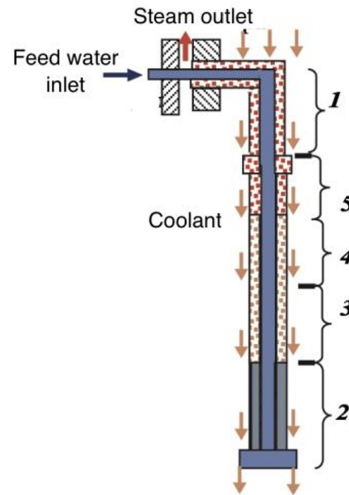


Figure 4.1:

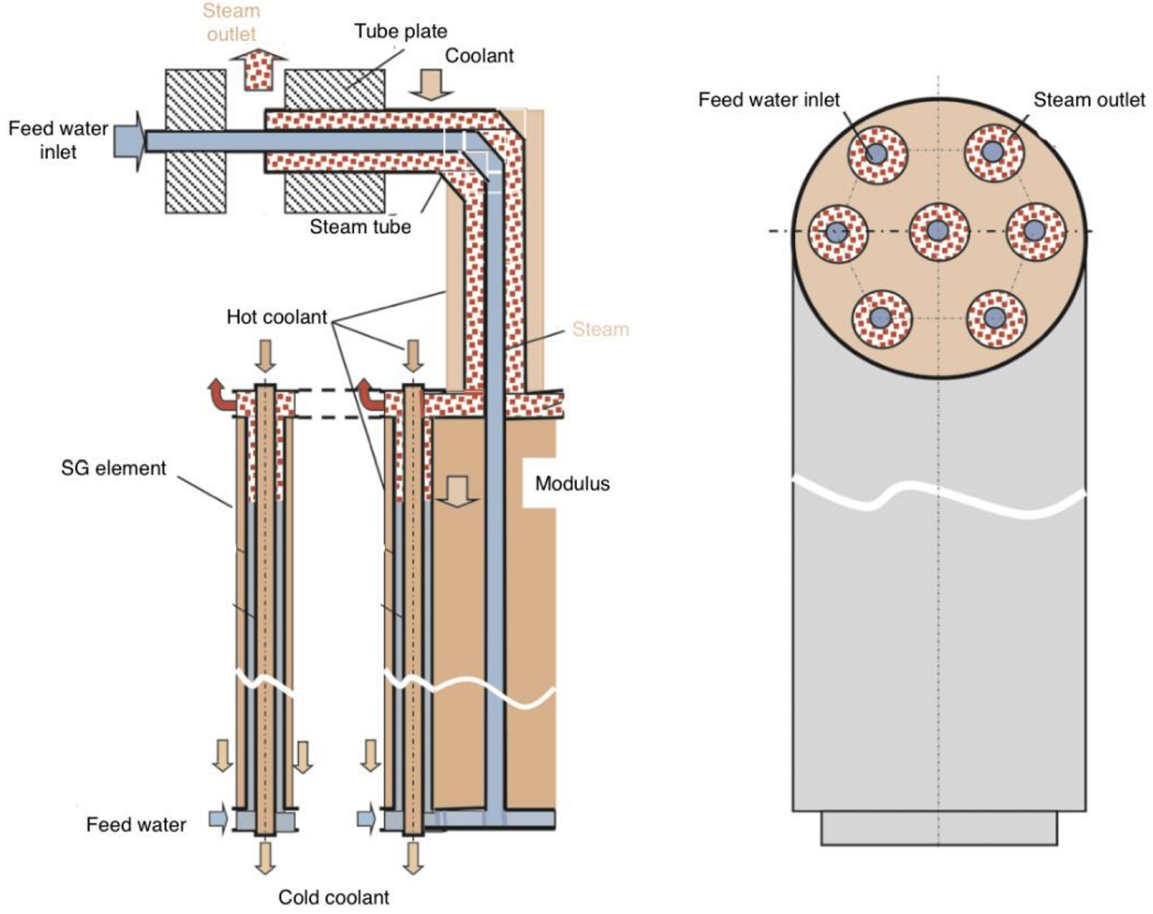


Figure 4.2: Subdivision of the fuel element

The cylindrical symmetry of the three regions allows for two fundamental simplifications, on which the entire model is based:

1. **Dimensional reduction (3D $\rightarrow$ 1D).** Thanks to the cylindrical symmetry, all quantities of interest (temperature, enthalpy, flow rate) can be averaged over each cross-section and treated as functions of the axial coordinate $z \in [0, L]$ alone, where L is the active length of the exchanger. The problem, originally three-dimensional, is thus reduced to a system of ordinary differential equations (ODEs) along z .
2. **Conduction resistances in cylindrical geometry.** Thermal conduction through the tube walls (of finite thickness) is described by the classical expression for the conductive resistance of a cylindrical shell of length dz :

$$R_{\text{cond}} = \frac{\ln(r_{\text{ext}}/r_{\text{int}})}{2\pi k dz} \quad (4.1)$$

where r_{int} and r_{ext} are the inner and outer radii of the tube and k is the thermal conductivity of the material. This expression, valid exclusively in cylindrical geometry, is applied to both the inner-tube wall and the outer-tube wall. The geometry is characterized by the diameters $(D_{1,\text{int}}, D_{1,\text{ext}}) = (5, 8)$ mm for the

inner tube and $(D_{2,int}, D_{2,ext}) = (10, 13)$ mm for the outer tube, with a triangular lattice pitch equal to:

$$P = 1.09 \cdot D_{2,ext}$$

Further assumptions relate to the constancy of material properties and to the neglect of fluid volume variations.

The system is discretized into N finite-volume cells of length $dz = L/N$, each treated as a control volume with uniform properties. The two fluids flow in **counter-current**: the primary enters at $z = L$ (hot) and exits at $z = 0$ (cold); the secondary enters at $z = 0$ (subcooled) and exits at $z = L$ (superheated steam). This configuration, chosen to maximize heat transfer, imposes opposite directions of advancement for the two fluids in the numerical solution.

Finally, the actual plant consists of a large number of identical sub-channels arranged in parallel (n_{sub} , given by the product of the number of groups, tubes per group, and sections); the model simulates a single sub-channel, assuming that the total flow rate is evenly distributed among all the channels, where the generic sub-channel was discretized into $N = 1000$ cells of equal length $\Delta z = L/N$, with $L = 1.9$ m, following a finite-volume scheme. The total number of parallel sub-channels for the whole system is $n_{sub} = 118 \times 7 \times 12 = 9912$. The total primary flow rate (flowrate I = 902.8 kg/s) and secondary flow rate (flowrate II = 77.5 kg/s) are evenly distributed among the sub-channels, defining the local flow rates $\dot{m}_I$ and $\dot{m}_{ann}$. The primary-side cross-section is further split between the inner area and the shell area:

$$A_{int} = \pi \left(\frac{D_{1,int}}{2} \right)^2, \quad (4.2)$$

$$A_{mant} = \frac{\sqrt{3}}{2} (1.09 \cdot D_{2,ext})^2 - \frac{\pi D_{2,ext}^2}{4} \quad (4.3)$$

proportionally to the respective flow cross-sectional areas. The geometry of the entire steam generator and of the various sections can be seen in Figures 4.1 and 4.2.

4.3 Thermal resistances and convective correlations

For each axial cell, heat exchange between the primary and secondary sides occurs through a path of resistances in series: primary-side convection $\rightarrow$ conduction through the tube wall $\rightarrow$ secondary-side convection. Since the primary fluid is in contact with both the inner tube and the outer tube, there are *two* parallel exchange paths toward the *annulus*, shown schematically in Figure 4.3.

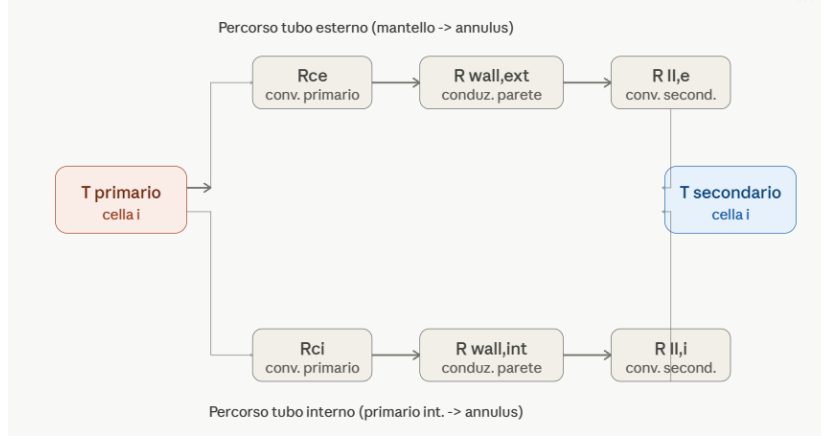


Figure 4.3: Series thermal-resistance diagram for a single cell, along the two paths (inner tube and outer tube)

4.3.1 Convective heat-transfer coefficients

The convective heat-transfer coefficient h is estimated using the **Dittus–Boelter** correlation:

$$Nu = 0.023 Re^{0.8} Pr^{0.4}, \quad h = \frac{Nu \cdot k_w}{D_h} \quad (4.4)$$

with $Re = \rho v D_h / \mu$ and $Pr = \mu c_p / k_w$, valid for fully developed single-phase turbulent flow.

For the **inner tube** and the **annulus**, where the flow cross-section is simple (circle or circular ring), Eq. (4.4) is applied directly with the geometric hydraulic diameter ($D_h = D$ for the tube, $D_h = D_{2,int} - D_{1,ext}$ for the annulus).

For the **shell**, on the other hand, the presence of the tubes arranged in a triangular pitch modifies both the flow area and the hydraulic diameter:

$$A_{\text{flow}} = \frac{\sqrt{3}}{2} P^2 - \frac{\pi D_{\text{ext}}^2}{4}, \quad D_h = D_{\text{ext}} \left(\frac{2\sqrt{3}}{\pi} \left(\frac{P}{D_{\text{ext}}} \right)^2 - 1 \right) \quad (4.5)$$

where P is the triangular pitch between the centers of adjacent tubes. The Nusselt number is further corrected with an empirical geometric factor, known as the Wiesman correction:

$$Nu_{\text{mantello}} = 0.023 Re^{0.8} Pr^{0.4} \left(1.106 \frac{P}{D_{\text{ext}}} - 0.013 \right) \quad (4.6)$$

which accounts for the increased turbulence induced by the triangular arrangement of the tubes relative to the case of an isolated tube.

4.3.2 Treatment of the secondary-side phases

As it crosses the exchanger, the secondary fluid goes from subcooled liquid to superheated steam, passing through a two-phase region. The local phase is determined from the cell's enthalpy h , by comparing it with the saturation values $h_{L,\text{sat}}$ and $h_{V,\text{sat}}$ at

constant pressure P_{II} :

$$\text{fase} = \begin{cases} \text{liquido} & h \leq h_{L,\text{sat}} \\ \text{bifase} & h_{L,\text{sat}} < h < h_{V,\text{sat}} \\ \text{vapore} & h \geq h_{V,\text{sat}} \end{cases} \quad (4.7)$$

In the two-phase region, the steam quality x is computed using the lever rule applied to enthalpy:

$$x = \frac{h - h_{L,\text{sat}}}{h_{V,\text{sat}} - h_{L,\text{sat}}} \quad (4.8)$$

and the temperature is set equal to the saturation temperature $T_{\text{sat}}(P_{II})$, while in the other two regions the temperature is obtained directly from the steam tables as $T = T(P_{II}, h)$.

The secondary-side convective coefficient depends on the phase:

- in the liquid or vapor phase, the correlations provided by pyXsteam are used, with the properties evaluated, respectively, from the fluid's temperature and pressure.
- In the saturated two-phase region, a semi-analytical formulation for convective boiling is adopted, based on the thermodynamic steam quality x , where the correlation used is the empirical one given by the Saha correlation. To avoid numerical discontinuities and to model the dryout phenomenon, the heat-transfer coefficient H is split into two regimes:

$$h_{\text{bif}} = h_l \left(1 + 3.8 \left(\frac{x}{1-x} \right)^{0.76} \left(\frac{3.4}{p_{\text{crit}}} \right)^{0.38} \right), \quad x < 0.8 \quad (4.9)$$

with $h_l = 6000$, while for $x \geq 0.8$ (post-dryout / mist-flow regime): a continuous linear interpolation is performed between the peak value computed at $x = 0.8$ (H_{08}) and the convective coefficient of superheated steam ($H_{II,v}$):

$$H = H_{08} + (H_{II,v} - H_{08}) \cdot \frac{x - 0.8}{1.0 - 0.8}$$

4.3.3 Total cell resistance

Combining primary-side convection, wall conduction (Eq. 4.1), and secondary-side convection, two functions were built for the total resistance along each of the two paths (inner tube i , outer tube e), where the resistance values for the k -th cell are:

$$R_i = R_{c,i} + R_{\text{wall,int}} + \frac{1}{2\pi r_{1,\text{ext}} dz h_{II}}, \quad R_e = R_{c,e} + R_{\text{wall,ext}} + \frac{1}{2\pi r_{2,\text{int}} dz h_{II}} \quad (4.10)$$

where the general R_{wall} values, again taken from (Eq. 4.1), are given by:

$$R_{\text{wall},i} = \frac{\ln(r_{i,\text{ext}}/r_{i,\text{int}})}{2\pi \cdot \Delta z \cdot k_{\text{cond}}} \quad (4.11)$$

with a wall thermal conductivity of $k_{cond} = 12 \text{ W m}^{-1} \text{ K}^{-1}$, while the generic $R_{c,i}$ and $R_{c,e}$ terms are computed at every time step based on the respective local heat-transfer coefficients and on the surface areas of the cell, using:

$$\frac{1}{2\pi \cdot (D_{i,int}/2) \cdot \Delta z \cdot H_{int}} \quad (4.12)$$

The following equations will also make use of the cell's overall thermal conductance, which is the sum of the two parallel conductances:

$$G = \frac{1}{R_i} + \frac{1}{R_e} \quad (4.13)$$

4.4 Discrete equations and time-advancement scheme

4.4.1 Energy balances on the control volume

For each cell, the energy balance is written in finite-volume form. For the primary fluid in cell i , the accumulated energy changes due to the incoming/outgoing advective flux and the heat given up to the secondary side:

$$\rho_I A_{\text{tot},I} dz c_{p,I} \frac{T_i^{n+1} - T_i^n}{dt} = \dot{m}_I c_{p,I} (T_{i+1}^{n+1} - T_i^{n+1}) - G_i (T_i^{n+1} - T_{II,i}) \quad (4.14)$$

where the advective term is written as an *upwind* flux with respect to the primary flow direction (from $i + 1$ toward i , since the primary advances from $z = L$ to $z = 0$). Similarly, for the secondary side in cell i :

$$\rho_{s,i} A_{\text{ann}} dz \frac{h_i^{n+1} - h_i^n}{dt} = \dot{m}_{\text{ann}} (h_{i-1}^{n+1} - h_i^{n+1}) + \frac{Q_i}{1000} \quad (4.15)$$

with an *upwind* flux from $i - 1$ toward i (the secondary advances from $z = 0$ to $z = L$) and $Q_i = G_i (T_i^{n+1} - T_{II,i})$ the heat exchanged in the cell (the factor 1000 converts from W to kW, since h is expressed in kJ kg^{-1}).

4.4.2 Implicit time discretization

Time advancement is carried out with an **implicit Euler** (*backward*) scheme. This choice is motivated by stability: an explicit scheme would be constrained by the *CFL* condition, $dt_{\text{max}} \approx dz/v$, which, for the fine axial discretization adopted, imposes time steps on the order of 10^{-3} s , making it computationally expensive to simulate transients lasting tens of seconds. The implicit scheme, by contrast, is unconditionally stable, allowing for time steps that are orders of magnitude larger with no onset of numerical oscillations.

Defining the accumulation terms (analogous to an equivalent cell mass flow rate):

$$\alpha = \frac{\rho_I A_{\text{tot},I} dz}{dt}, \quad \beta_i = \frac{\rho_{s,i} A_{\text{ann}} dz}{dt} \quad (4.16)$$

Eqs. (4.14)–(4.15), solved explicitly for the unknown at step $n + 1$, become:

$$T_i^{n+1} = \frac{\alpha T_i^n + \dot{m}_I T_{i+1}^{n+1} + \frac{G_i}{c_{p,I}} T_{II,i}}{\alpha + \dot{m}_I + \frac{G_i}{c_{p,I}}} \quad (4.17)$$

$$h_i^{n+1} = \frac{\beta_i h_i^n + \dot{m}_{\text{ann}} h_{i-1}^{n+1} + \frac{\delta Q_i}{1000}}{\beta_i + \dot{m}_{\text{ann}}} \quad (4.18)$$

where the semi-implicit thermal source term is expressed by:

$$\delta Q_i = \max(0, T_I^{n+1}[i] - T_{II}^n[i]) \cdot G_i$$

This is a system of implicit equations, since T_{i+1}^{n+1} and h_{i-1}^{n+1} themselves appear as unknowns. However, thanks to the one-directional structure of the upwind advection, the system requires no matrix inversion: it can be solved by **sequential substitution** (*sweep*), following the natural order imposed by the flow direction:

- the **primary** side is updated by sweeping from $i = N - 1$ to $i = 0$: cell $N - 1$ uses $T_{I,\text{in}}$ as the boundary condition, and each subsequent cell uses the just-computed value T_{i+1}^{n+1} ;
- the **secondary** side is updated by sweeping from $i = 0$ to $i = N - 1$: cell 0 uses $h_{II,\text{in}}$ as the boundary condition, and each subsequent cell uses the just-computed h_{i-1}^{n+1} .

The exchange term $G_i(T_i - T_{II,i})$ is treated in a **semi-implicit** way: the secondary temperature $T_{II,i}$ used in updating the primary side (Eq. 4.17) is the one known from the previous step n , while the flux Q_i used in updating the secondary side (Eq. 4.18) employs the primary temperature T_i^{n+1} already updated within the same step. This sequential coupling (primary $\rightarrow$ secondary) avoids the solution of a full $2N \times 2N$ linear system, while still maintaining good stability thanks to the implicit treatment of the dominant advective terms.

An additional physical constraint imposes that heat cannot flow from the secondary side to the primary side: if the implicit update would return $T_i^{n+1} < T_{II,i}$, the cell is recomputed by zeroing the exchange term,

$$T_i^{n+1} = \frac{\alpha T_i^n + \dot{m}_I T_{i+1}^{n+1}}{\alpha + \dot{m}_I} \quad (4.19)$$

and, similarly, the flux Q_i in Eq. (4.18) is limited by $\max(0, T_i - T_{II,i})$.

The thermodynamic state, the temperature T_{II} , and the secondary-side steam quality x are updated cell by cell by interfacing the enthalpy h^{n+1} with the pyXSteam library routines at the operating pressure $P_{II} = 34$ bar, distinguishing between the subcooled-liquid ($h \leq h_L^{\text{sat}}$), saturated two-phase ($h_L^{\text{sat}} < h < h_V^{\text{sat}}$), and superheated-steam ($h \geq h_V^{\text{sat}}$) regimes.

4.5 Solution strategy

The complete solution is organized into three successive phases, each built on the result of the previous one.

4.5.1 Continuous steady state: shooting method

The first phase determines an estimate of the steady-state solution using a **shooting method**. The steady-state problem, obtained by setting the time derivatives to zero in Eqs. (4.14)–(4.15), is a boundary-value problem (the primary temperature is known at $z = L$, i.e. $T_{p,i} = 313,5^\circ\text{C}$, and the secondary temperature is also known at $z = 0$, i.e. $T_{s,i} = 170^\circ\text{C}$, but they must be propagated in opposite directions), where the system's boundary conditions are intrinsically coupled at opposite ends because of the counter-current configuration. An algorithm is therefore used that guesses, by trial, the primary outlet temperature $T_{I,\text{out}}$ (at $z = 0$): starting from a trial value, the steady-state ODE system is integrated

$$\frac{dT_I}{dz} = \frac{G(T_I - T_{II})}{\dot{m}_I c_{p,I}}, \quad \frac{dh_{II}}{dz} = \frac{G(T_I - T_{II})}{\dot{m}_{\text{ann}}} \quad (4.20)$$

using an explicit upwind scheme from $z = 0$ to $z = L$, and it is checked whether the primary temperature integrated at $z = L$ matches the known boundary condition $T_{I,\text{in}}$. The correct initial condition is found by minimizing the residual with respect to the actual imposed inlet temperature ($T_{I,\text{in}} = 315.5^\circ\text{C}$) using Brent's method (*root scalar*). The continuous profile thus obtained is projected onto the spatial grid through a geometric offset (*face-to-center correction*) for the secondary enthalpy, and is used as input for an implicit warmup process with $\Delta t = 1$ s. This numerical double-step allows convergence to the discrete steady state in fewer than 300 iterations, ensuring a perfect trilateral energy balance (with a numerical imbalance among the primary, secondary, and wall fluxes of less than 0.001%) before triggering the dynamic transient.

4.5.2 Implicit warmup toward the discrete steady state

The solution obtained from the shooting method is an accurate solution of the *continuous* problem, but it does not exactly coincide with the steady-state solution of the *discretized* system with N finite cells (the continuous and discrete *upwind* schemes introduce slightly different truncation errors). To obtain an initial condition consistent with the numerical scheme that will later be used in the transient, the *shooting* profile is evolved through the implicit scheme (Eq. 4.17–4.18) with a large time step ($dt = 1$ s), until convergence. Convergence is declared when the maximum variation in temperature and enthalpy between two successive steps remains below tolerance for a preset number of consecutive iterations, avoiding false positives due to transient numerical fluctuations.

At the end of the *warmup*, the consistency of the solution is verified by comparing three independent estimates of the total heat exchanged: the cell-by-cell sum of the local fluxes $Q = \sum_i G_i(T_i - T_{II,i})$, the energy balance on the primary side $Q = \dot{m}_I c_{p,I}(T_{I,\text{in}} - T_{I,\text{out}})$, and the balance on the secondary side $Q = \dot{m}_{\text{ann}}(h_{\text{out}} - h_{\text{in}})$.

The agreement (up to minimal numerical error) among the three estimates constitutes an independent check of the correctness of the implementation.

4.5.3 Transient: secondary flow-rate step

Starting from the discrete steady state, an **increase or decrease, with saturating growth**, of the secondary mass flow rate is simulated at time $t = 10$ s, keeping all other boundary conditions unchanged. This was done in order to simulate the actual transient carried out by a pump in the secondary system. The integration proceeds with the implicit scheme (Eq. 4.17–4.18) with a time step $dt = 0.1$ s up to $t = 30$ s, sufficient to observe both the transient immediately following the step and the attainment of a new equilibrium state. At every step, the primary-side convective resistances (Eq. 4.4–4.6) are recomputed, since they depend on the flow rate through the Reynolds number, and therefore vary in time in response to the step.

The quantities of interest – primary outlet temperature and exchanged thermal power (computed with the three independent methods) – are saved at every time step. Comparing the average values in the window immediately preceding the step with those in the final window of the simulation gives the net variation induced by the transient on the output quantities.

Chapter 5

Pressure Evolution

The secondary circuit includes a turbine that regulates the outgoing steam flow rate, which in turn regulates the extracted power. This directly affects the thermodynamic conditions inside the heat exchanger. An increase in the flow rate demanded by the turbine leads to a lower total steam volume, and hence to a decrease in pressure on the secondary side. This pressure reduction favors greater heat transfer from the primary circuit to the secondary circuit, increasing the steam generation rate. Conversely, a decrease in the demanded flow rate reduces the steam draw-off, causing an increase in pressure on the secondary side of the steam generator and a consequent reduction of the thermal gradient available for heat exchange. The pressure in the steam generator is therefore closely correlated with the secondary fluid flow rate and represents one of the main controlled variables for ensuring correct energy balancing and the operational stability of the plant.

5.1 Approximate model

For the calculation of the pressure variation over time, an approximation of the real physical problem was adopted through the introduction of the constants K_p and τ_p , which represent, respectively, the sensitivity of pressure to flow rate and the time constant. The secondary fluid is always considered to be in saturation conditions, which is true when it is in the two-phase state and is a good approximation for the outlet conditions. The subcooled-liquid zone occupies a limited fraction of the steam generator's length, since the enthalpy jump required to bring the fluid to saturation is significantly smaller than the latent heat of vaporization. The assumption $u \approx u_{sat}(P_s)$ introduced in the pressure balance therefore introduces an error limited to the secondary inlet zone alone, and is considered acceptable as a first-order approximation.

The coefficient K_p of the phenomenological model can be derived from the internal-energy balance on the secondary-side control volume. Denoting by $U = M_s u_{sat}(P_s)$ the total internal energy:

$$\frac{dU}{dt} = \dot{Q}_{in} - \dot{m}_{II} \Delta h \quad (5.1)$$

Since the latter depends on pressure, which in turn depends on time, we have:

$$\frac{dP_s}{dt} = \frac{\dot{Q}_{in} - \dot{m}_{II} \Delta h}{M_s \frac{du_{sat}}{dP_s}} \quad (5.2)$$

Close to the steady-state regime, where $\dot{Q}_{in,0} = \dot{m}_{II,0} \Delta h_0$, a perturbation $\delta \dot{m}$ of the secondary flow rate produces:

$$\frac{dP_s}{dt} \approx -\frac{\Delta h_0}{M_s \frac{du_{sat}}{dP_s}} \delta \dot{m} = -K_p \delta \dot{m} \quad (5.3)$$

The coefficient K_p is therefore:

$$K_p = \frac{\Delta h_0}{M_s \frac{du_{sat}}{dP_s}} \quad [\text{bar}/(\text{kg/s/s})] \quad (5.4)$$

The relaxation term $-(P_s - P_{s,0})/\tau_p$ in the phenomenological model instead represents the stabilizing effect of the thermal feedback: as P_s decreases, T_{sat} falls, $\dot{Q}_{in}$ increases, and the pressure tends to stabilize at the new equilibrium value. The constant τ_p does not emerge directly from the physical balance but parametrizes the speed of this feedback, which in the physical model (5.2) is instead captured implicitly through the dependence of $\dot{Q}_{in}$ on P_s via $T_{sat}(P_s)$.

5.2 Results

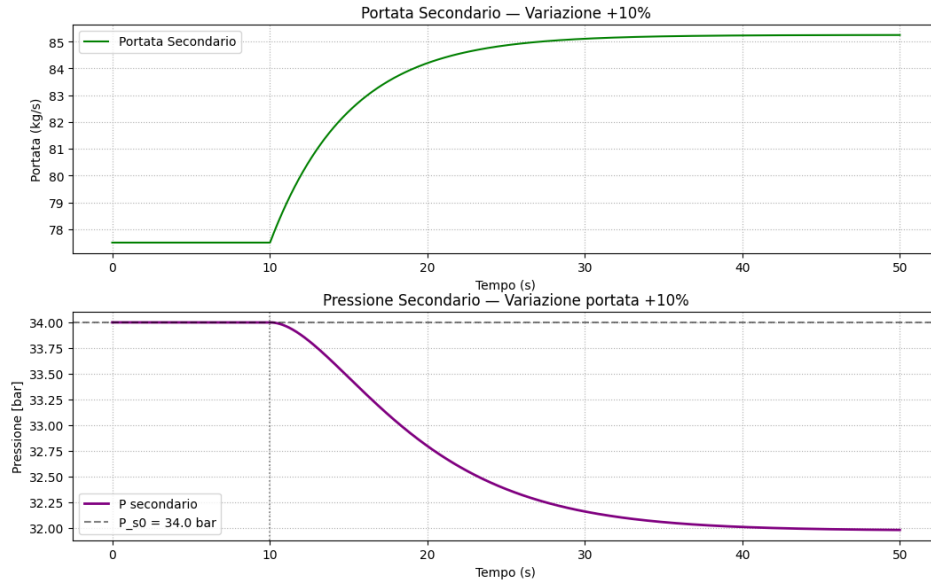


Figure 5.1: Time variation of the secondary-side pressure, due to a +10% increase in secondary flow rate

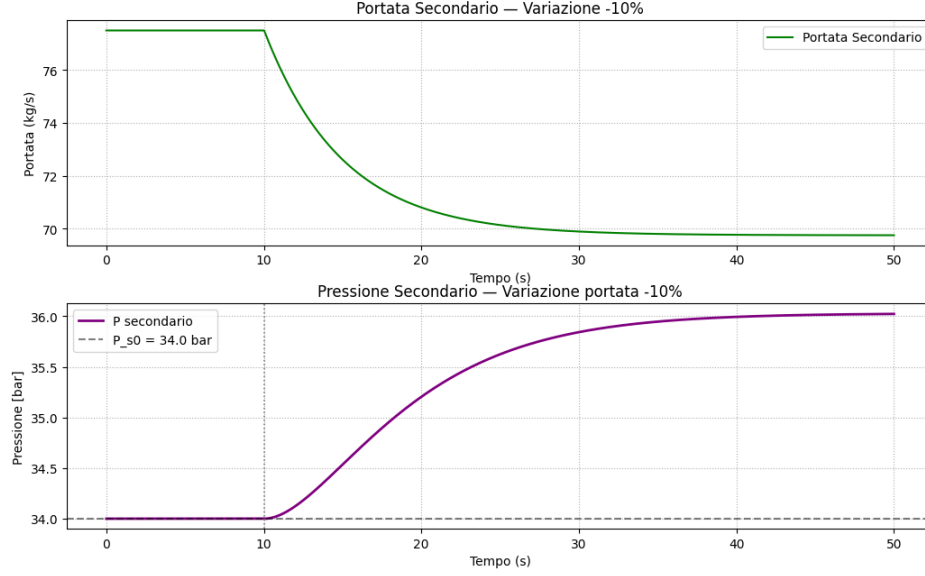


Figure 5.2: Time variation of the secondary-side pressure, due to a -10% increase in secondary flow rate

The pressure time constant was set equal to that of the flow rate: $\tau_p = \tau_{fr} = 5$ s. It is important to note that the apparent response time of the pressure in the plots is larger than τ_p : this is because the instantaneous equilibrium point is not fixed but shifts over time together with the flow rate, which itself evolves with its own time constant. The pressure therefore chases a moving target, and the resulting dynamics produce a response that appears slower than τ_p .

The coefficient K_p , computed from the physical balance described above, turns out to be $K_p = 52.36 \times 10^{-3}$ bar/(kg/s/s).

Chapter 6

Analysis of the Complete System

The simulation of the complete system represents the final result of the work, in which the models developed in the previous chapters are integrated into a single coupled solution scheme. The goal is to faithfully reproduce the dynamic response of the RITM-200 reactor to operational perturbations, taking into account the interdependence of the neutronic, thermal-hydraulic, and thermodynamic phenomena that govern the system's behavior.

6.1 Coupling strategy

The main source of complexity in simulating the complete system lies in the coupling between the core model and the heat-exchanger model. The two subsystems are physically connected through the primary circuit: the core outlet temperature (*hot leg*) constitutes the inlet boundary condition for the *SG*, while the outlet temperature of the *SG* (*cold leg*) determines the coolant inlet temperature into the core, directly influencing the reactivity through the moderator feedback coefficient α_m .

This circular dependence – the core influences the *SG* and the *SG* in turn influences the core – makes the problem intrinsically coupled and requires a solution strategy that guarantees physical consistency between the two models at every time step. The adopted solution consists in integrating the core's ODE system within the *SG*'s time loop, so that the two models share the same time discretization. At each step Δt :

1. The *SG* is updated with the implicit finite-volume scheme (Eq. 4.17-4.18), using as the boundary condition the *hot leg* temperature computed at the previous step.
2. The updated *cold leg* temperature T_{cold}^{n+1} , extracted from the primary profile at the *SG* outlet, is passed to the core model as the inlet boundary condition.
3. The core's ODE system – comprising the point-kinetics equations and the three-radial-node thermal-hydraulic model – is integrated over the interval $[t^n, t^{n+1}]$ using `solve_ivp` with the LSODA method, yielding the updated temperatures of the fuel, the cladding, and the coolant, as well as the new neutron density n^{n+1} .
4. The reactivity is recomputed with the updated temperatures according to Eq. (3.21), closing the feedback loop.

This scheme can be classified as a sequential method: the two physical operators (transport in the *SG* and core kinetics) are solved in sequence within the same time step, passing each other the updated boundary conditions. The approach introduces a delay of one time step in the coupling, the effect of which on the accuracy of the solution is negligible for the time steps adopted ($\Delta t = 0.2$ s), significantly smaller than the time constants of the transients analyzed (of the order of tens of seconds).

6.2 Pressure model in the coupled system

In the complete system, the secondary-circuit pressure is no longer a fixed parameter but a dynamic variable, updated at every time step according to the model described in Chapter 5. The pressure variation changes the saturation properties of the secondary fluid (in particular the saturation temperature $T_{sat}(P_s)$ and the enthalpies h_L^{sat} , h_V^{sat}), which are therefore recomputed at every step using the `pyXSteam` tables. This introduces a further coupling between the secondary flow-rate dynamics, the pressure, and the heat exchange in the *SG*, making the system fully feedback-coupled.

6.3 Solution scheme

The overall solution scheme of the simulation can be summarized in the following phases:

1. **Initialization.** Declaration of the physical and geometric parameters of the core and of the *SG*; computation of the nominal thermodynamic properties using `pyXSteam`.
2. **Coupled steady state.** Iterative solution of the nonlinear algebraic system that couples the core's steady-state equations (solved with `fsolve`) and the shooting method for the *SG* (with `root_scalar`), until convergence of the *cold leg* temperature.
3. **Coupled warmup.** Implicit integration over 1000 steps with $\Delta t_{wu} = 1$ s, aimed at bringing the discrete profiles of the *SG* and of the core to the steady state consistent with the numerical scheme adopted, eliminating the numerical discontinuities due to the projection of the continuous profile onto the discrete grid.
4. **Transient simulation.** Coupled core-*SG* integration with $\Delta t = 0.2$ s for the total duration of the transient, with the secondary flow rate, the pressure, the convective resistances, and the reactivity updated at every step.
5. **Post-processing.** Saving and plotting of the variables of interest: neutron density, fuel and coolant temperatures, *hot leg* and *cold leg* temperatures, secondary-side outlet enthalpy and temperature, exchanged power, and secondary-side pressure.

Chapter 7

Final Considerations on the Results

The final results of the simulation clearly show the effects that deviations from the nominal secondary-circuit flow rate, and variations in control-rod height, produce on the entire system, comprising the primary-circuit coolant water and the secondary working fluid.

As a first case study, a single perturbation consisting of a 10% increase in the secondary-circuit flow rate is examined.

7.1 Flow-rate variation

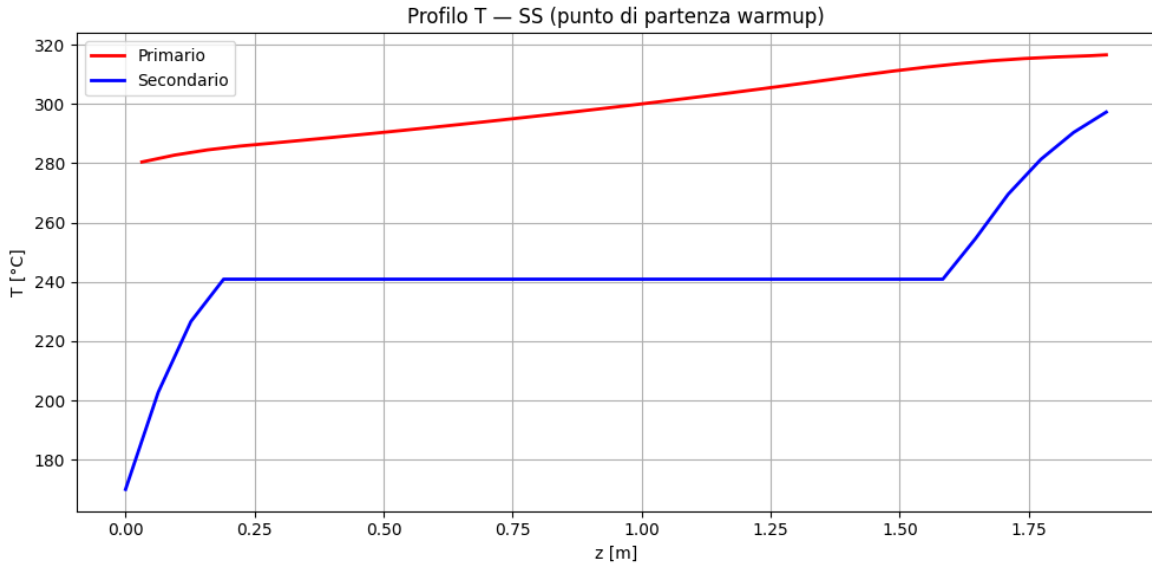


Figure 7.1: Steady state under nominal conditions, showing the primary and secondary temperature distribution along the length of the *SGE*

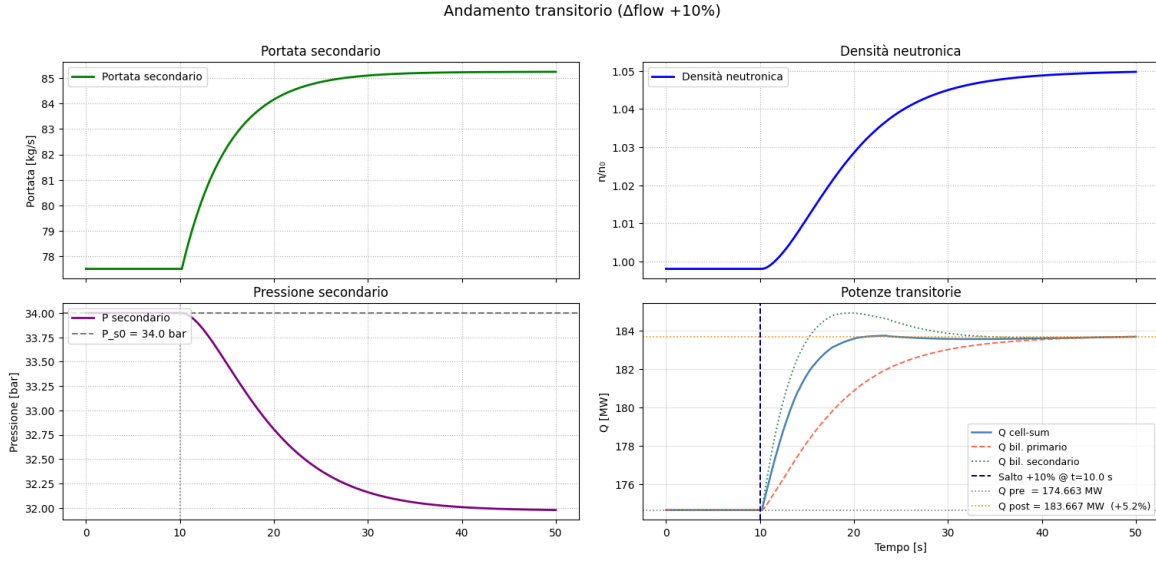


Figure 7.2: Time variation of: secondary flow rate, secondary pressure, neutron density, and exchanged thermal power

It can be observed that the trends of the variables (with the exception of pressure, for which a correlation with flow rate was imposed) show a similar simple exponential behavior. This happens as a result of a linear relationship between mass flow rate and, respectively, neutron density and exchanged thermal power:

$$\dot{Q} = \dot{m}(T_{out} - T_{in}); \quad \dot{Q} = \dot{Q}_0 \frac{n}{n_0} \quad (7.1)$$

The pressure, on the other hand, decreases as a consequence of the increase in steam extracted from the exchanger, resulting from the increase in secondary-circuit flow rate that feeds the turbine (and increases its generated power).

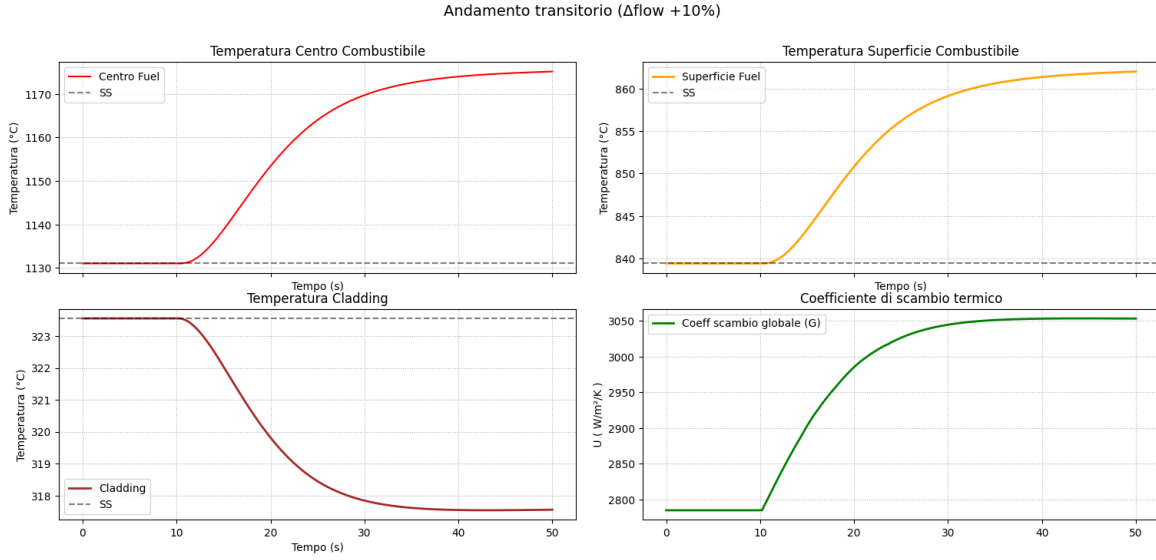


Figure 7.3: Time variation of: fuel centerline temperature, fuel surface temperature, overall heat-transfer coefficient, cladding temperature

The fuel temperatures rise consistently with the increase in neutron density; the heat-transfer coefficient, in turn, increases in line with the increase in secondary-circuit flow rate, and hence with the contribution of the *Reynolds number*; the cladding temperature decreases owing to the decrease in mean coolant temperature (as shown below).

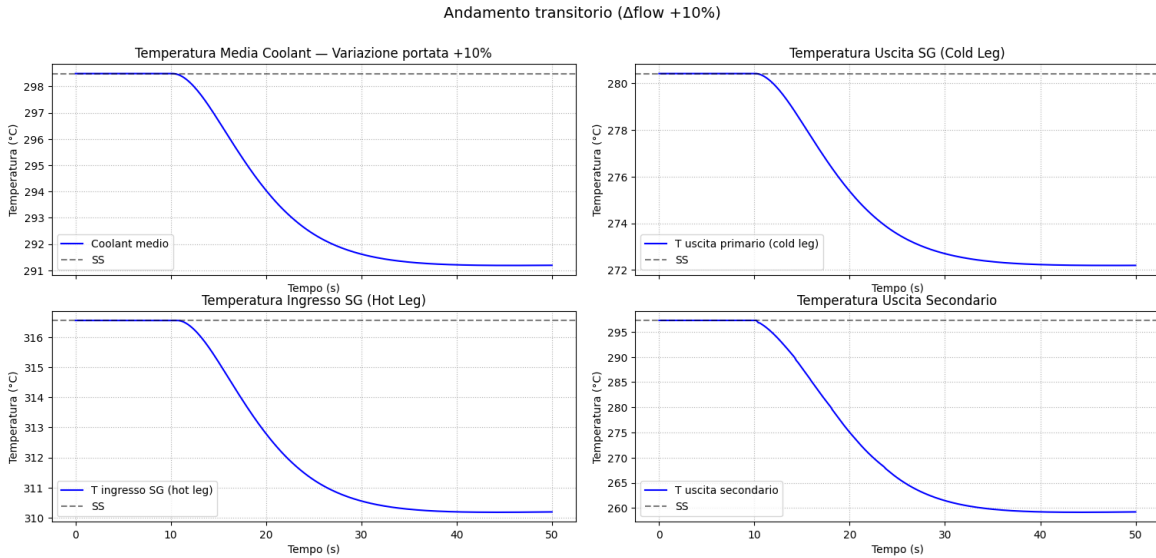


Figure 7.4: Time variation of: mean coolant temperature, *hot leg* temperature, *cold leg* temperature, secondary outlet temperature

The secondary-circuit steam temperature trend is decreasing, since the decrease in pressure leads to a lower saturation temperature and hence to less superheating of the

steam. The primary circuit follows exactly the trend of the secondary circuit, which removes more heat and therefore undergoes substantial cooling.

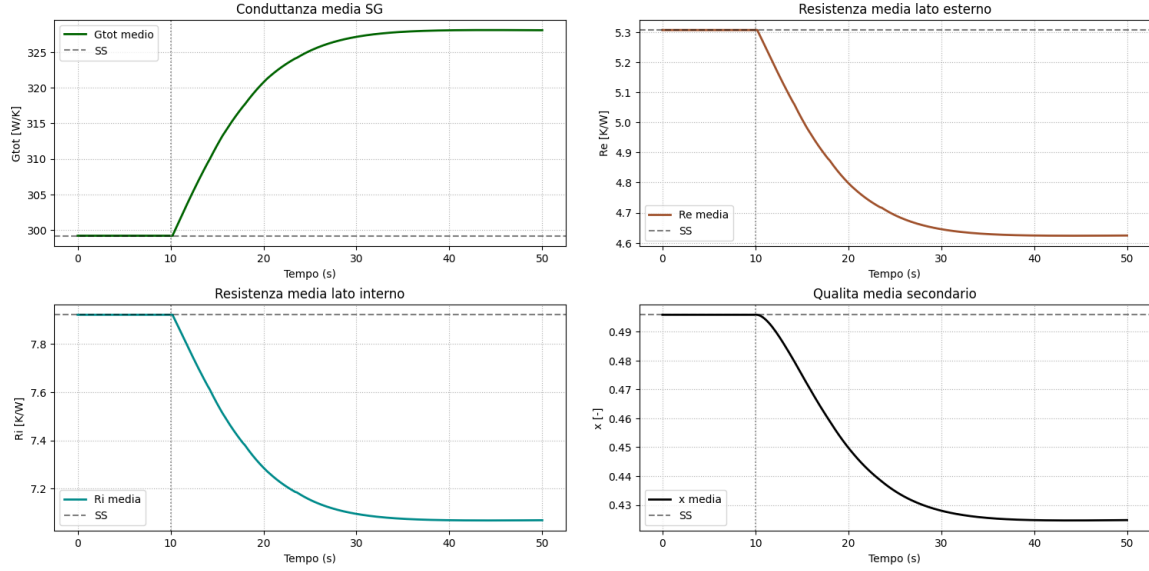


Figure 7.5: Time variation of: mean conductance, mean shell-side resistance, mean tube-side resistance, mean secondary-side steam quality

The trend of the first three parameters is a direct consequence of the increase in the heat-transfer coefficient, due to the increase in secondary flow rate. The mean quality, on the other hand, decreases (although not significantly), since the larger amount of steam extracted, in order to be sent to the turbine, allows a prevalence of the liquid phase in the *SGE* (the balance nevertheless settles on a steam mass not much lower than the liquid mass).

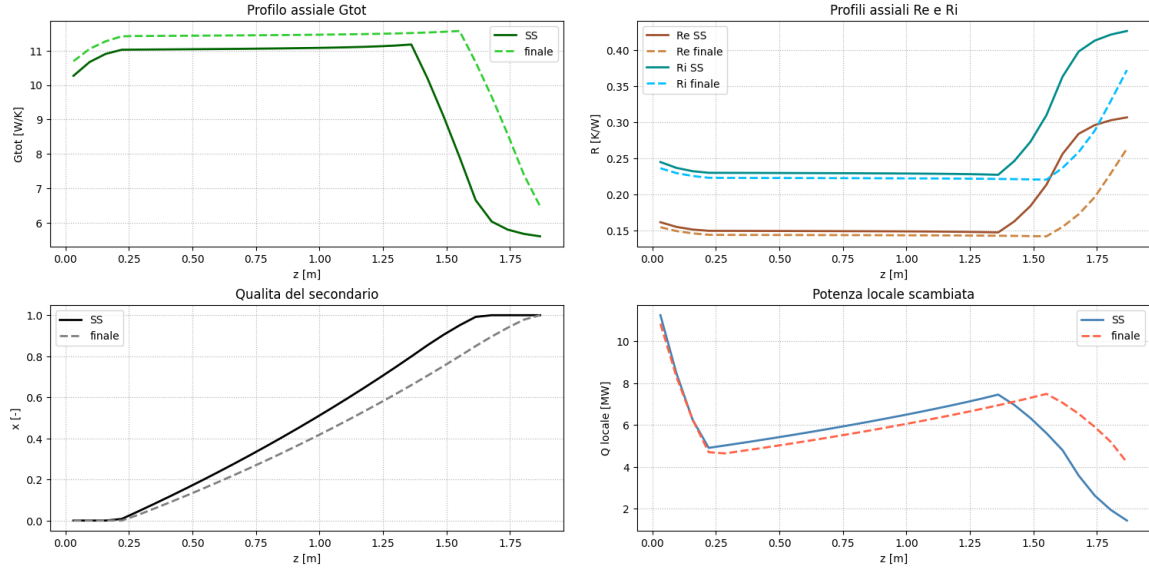


Figure 7.6: Axial profiles of: total conductance, shell-side resistance, tube-side resistance, steam quality, local exchanged power

The distribution of conductance along the *SGE* clearly shows the markedly higher propensity for heat exchange of the liquid and two-phase states, compared with vapor, which has a markedly lower conductance. This phenomenon can also be observed through the trends of the resistances; the situation does not change significantly in the final state, whose only difference is a larger fraction of two-phase vapor along the axis. Looking at the secondary-circuit quality profile in the final state, unity is reached only close to the very end; this indicates that the steam entering the turbine undergoes minimal superheating, unlike the initial state, in which the steam has more time to become superheated (since the flow rate is lower). Finally, in line with the heat-transfer coefficients and the temperature differences between the primary and secondary sides, the profiles (nearly unchanged between the final and initial states) of the power exchanged along the *SGE* can be appreciated: the liquid zone provides a very high heat exchange at the initial end, due mostly to the large temperature difference, which tends to decrease sharply as the temperatures converge; the profile rises again with the development of the two-phase state along the axis; it ends with a collapse in heat exchange along the vapor zone (as expected, given the low heat-transfer coefficient). It is interesting to note how the two-phase state, thanks to the nucleate-boiling phenomenon, keeps the power exchanged along the axis high (unlike the liquid state), despite the progressive convergence of the temperatures. This is a consequence of the extraordinary heat-exchange capacity of boiling fluids.

7.2 Logical Scheme of the Phenomena Analyzed in the Plots (1)

The main objective of the simulation is to understand clearly the dynamic behavior of the reactor following a possible operational perturbation. To achieve this, it is essential

to take into account the causal relationship between the phenomena that occur within the system. The increase in reactor power is justified by an increase in the energy demand from the electrical grid, which falls directly on the turbine downstream of the secondary side. It is at this point that, by means of a valve, the steam flow rate entering the turbine is regulated; for an increase in power, the flow rate must clearly increase (e.g. by +10%).

The higher flow rate causes an increase in the heat removed by the secondary circuit inside the *SG*, and in parallel a decrease in pressure (controlled by the amount of steam present in the *SG*) on the secondary side; this pressure decrease is controlled by a correlation assumed to be linear with the variation of the flow rate itself. The increase in flow rate, and hence in the secondary-side transit velocity, is responsible for the general increase in the heat-transfer coefficient; furthermore, the decrease in pressure allows the secondary side to settle at a lower saturation temperature, which increases the temperature difference with the primary side. The increases in ΔT and U allow the primary circuit to supply the greater thermal power demanded by the secondary side.

In the primary circuit, from which more heat is removed, a temperature drop follows, similar to that of the secondary circuit, settling at an *SG* outlet temperature (i.e. *cold leg*) lower than the nominal one. A decrease in *cold leg* temperature results in colder, and hence denser, water entering the core; given the dual role of coolant and moderator played by the primary water, an increase in density allows for better moderation of the neutron activity and hence a positive reactivity insertion within the core.

This highly significant phenomenon is the reason why *PWR*-type reactors are called *load followers*, i.e., capable of automatically compensating for the higher power demand from the turbine, through a natural increase in core reactivity, which inevitably develops a higher thermal power. All of this happens almost simultaneously with the thermal-hydraulic phenomena. This is evident from the fact that, within 50 seconds, all the analyzed variables end up converging to the new state.

Finally, despite the increase in thermal power in the core, the *hot leg* temperature (entering the *SG*) stabilizes at a temperature lower than the previous one, causing a decrease in the mean primary temperature, while the fuel temperature increases in line with the neutron density.

7.3 Control-Rod Variation

It is possible to analyze the behavior resulting from a variation in control-rod height. For example, a 3 cm insertion of rods into the *core*. The steady-state profiles are practically unchanged

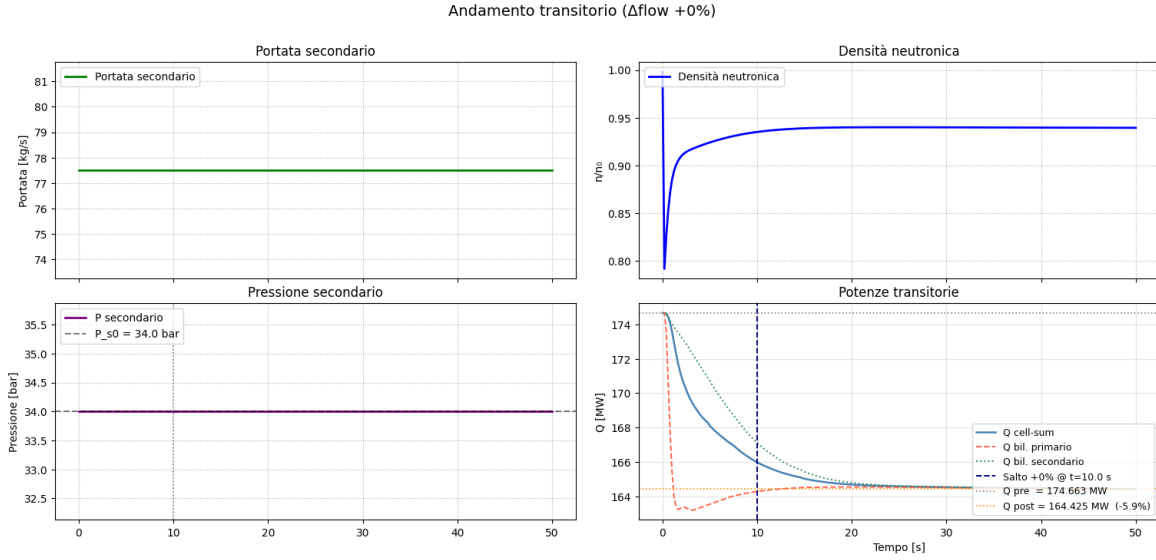


Figure 7.7: Time variation of: secondary flow rate, secondary pressure, neutron density, and exchanged thermal power

The secondary flow rate, and hence the pressure, remain unchanged. The neutron density, on the other hand, decreases following a specific pattern: the initial drop is due to the immediate absorption of neutrons by the control rods, while the slight rebound, which settles at a value lower than the initial one, is due to the cooling of the primary-circuit water, with a consequent increase in its density and moderating capacity.

The power decreases, since the drop in neutron density hinders the production of fission thermal power within the fuel.

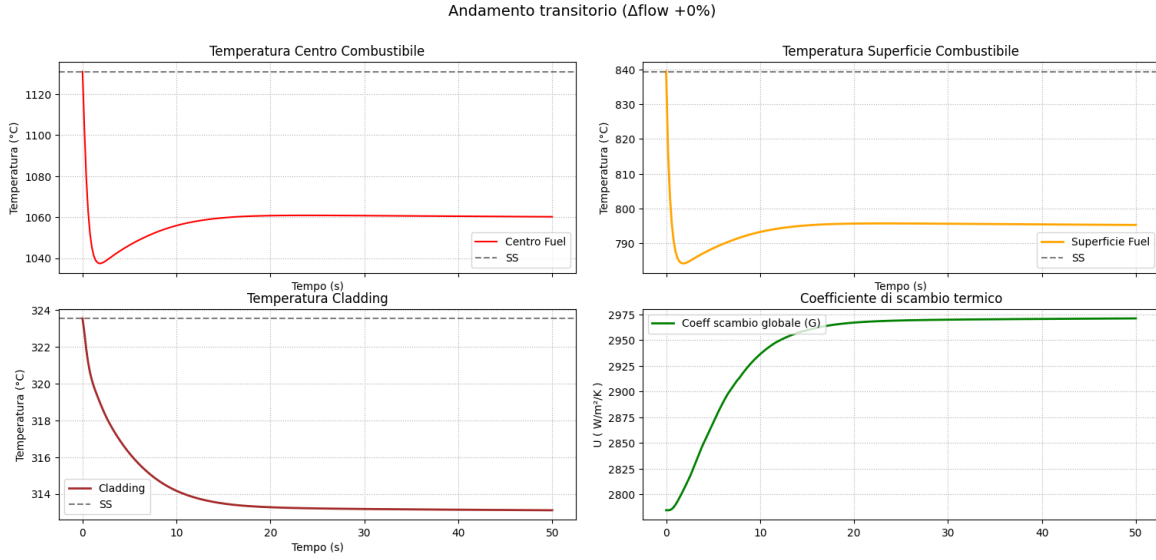


Figure 7.8: Time variation of: fuel centerline temperature, fuel surface temperature, overall heat-transfer coefficient, cladding temperature.

The fuel temperature (center and surface) follows a trend very similar to that of the neutron density, being a direct consequence of it. The cladding temperature, on the other hand, decreases as a result of the (previously introduced) cooling of the primary-circuit water.

It is interesting to note that, contrary to the decrease in exchanged thermal power, the heat-transfer coefficient between the primary and secondary circuits increases. This is due to the dominant presence of liquid and two-phase flow within the *SGE* (as will be explained below), which contributes to raising the quality of the heat exchange.

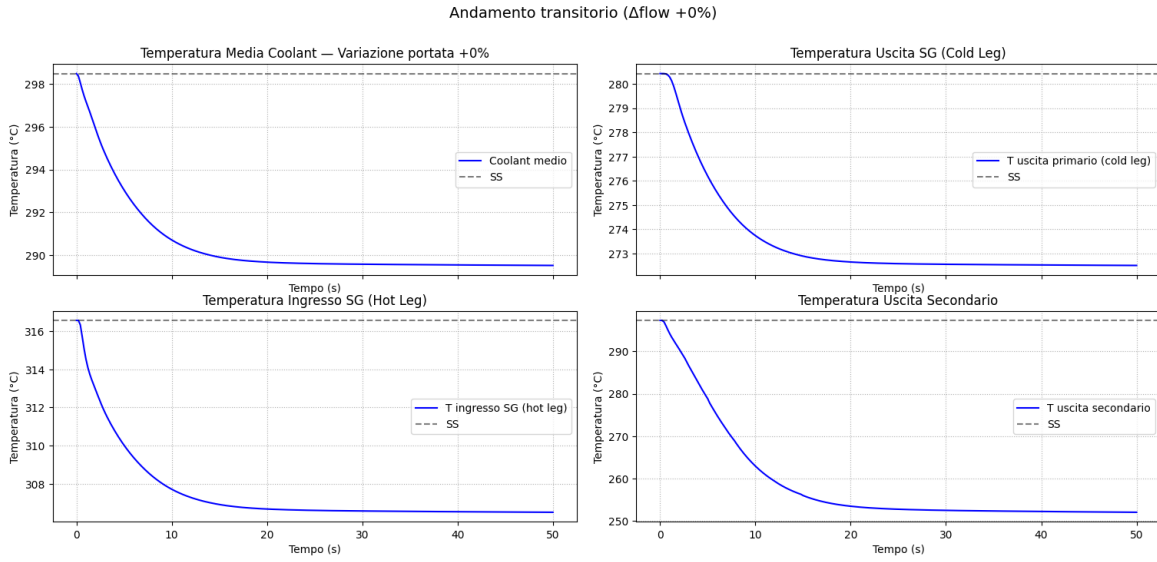


Figure 7.9: Time variation of: mean coolant temperature, *hot leg* temperature, *cold leg* temperature, secondary outlet temperature

The decrease in the primary water temperature is a consequence of the decrease in fuel temperature and of the lower thermal power absorbed. The cooling of the secondary circuit likewise depends on the power exchanged with the primary side, which drags the secondary side down with it (with the exception of the two-phase-zone stretch, which remains at the saturation temperature)

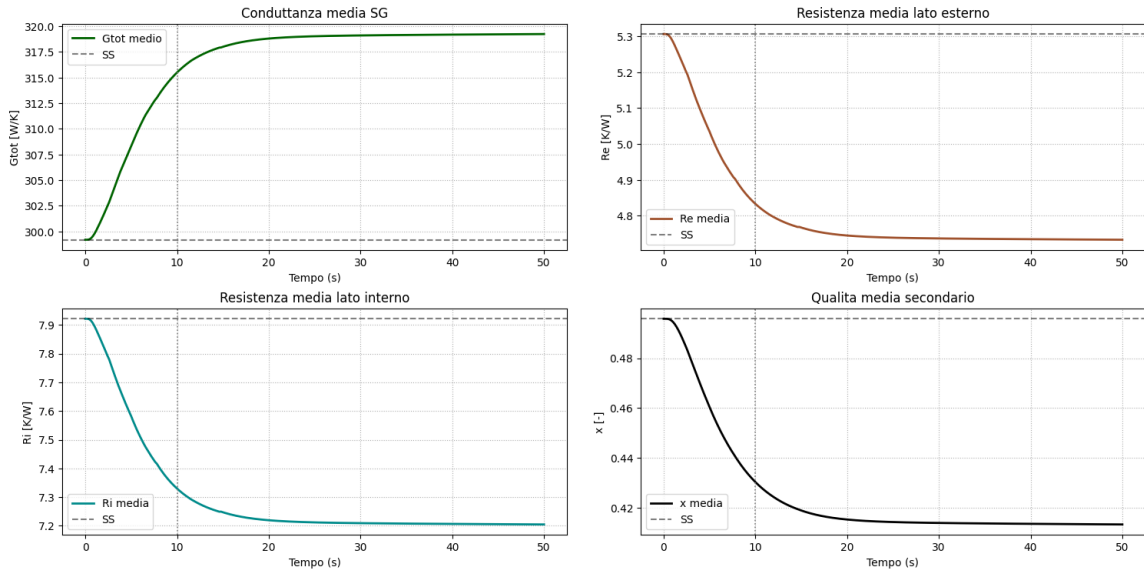


Figure 7.10: Time variation of: mean conductance, mean shell-side resistance, mean tube-side resistance, mean secondary-side steam quality.

The trends of conductance and resistance (shell side and tube side) are easily de-

duced from the previous considerations regarding the heat-transfer coefficient. The mean secondary-side quality decreases. This is due to the collapse in the amount of steam present in the *SGE*, caused by the general cooling, which leaves more room for the presence of liquid and two-phase flow.

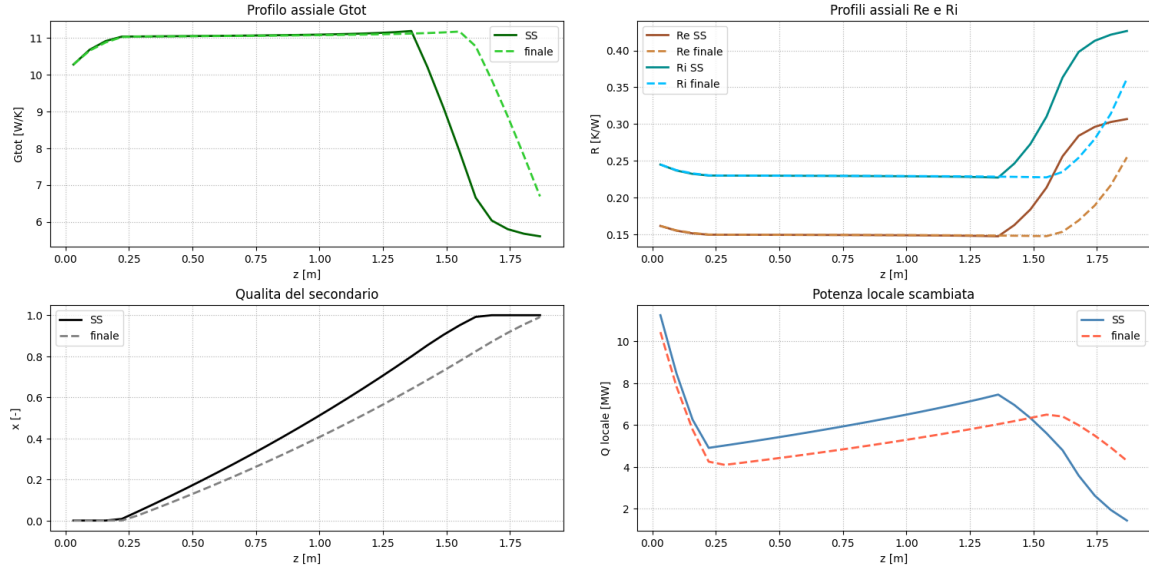


Figure 7.11: Axial profiles of: total conductance, shell-side resistance, tube-side resistance, steam quality, local exchanged power

The analysis of the profiles along the axis is of particular interest: the total conductance, as previously discussed, increases over time; indeed, it can be seen that in the final state it remains high for longer than in the initial state – this is due solely to the lack of thermal power entering the secondary-circuit water, which struggles more to evaporate, so that, over the same length of the *SGE*, there is a greater presence of two-phase flow. It is clear that the conductance of the two-phase region (in the central zone) and of the liquid region (in the left-hand zone) has no reason to vary between the two states, since the flow rate remains unaltered. A similar result is found for the resistances. As in the previous case, the secondary-side quality reaches unity close to the end of the *SGE* (although for different reasons). The power exchanged along the axis obviously decreases overall in the final state, although it remains high for longer owing to the persistence of the two-phase state.

7.4 Logical Scheme of the Phenomena Analyzed in the Plots (2)

In this scenario, the chain of events is reversed compared with the previous case. The main cause of the power decrease is in fact upstream, in the lowering of the control rods, aimed precisely at controlling the core power; it is therefore a direct action on the change of the reactor state, and not a consequence of the downstream turbine regulation as in the previous case. The insertion of the rods causes an immediate absorption of

neutrons in the core and hence an inhibition of the fission chain reaction, with a consequent drop in the generated power. This causes a decrease in the fuel temperature (a direct consequence), in the cladding temperature, and in that of the primary circuit (an effect of the lower power flowing through it). It is precisely the decrease in the primary water temperature, and the increase in its density, that allows the slight rebound in neutron density. The interesting paradox is found in the *SG*, in the thermal interaction between the primary and secondary sides. Essentially, the heat to be exchanged decreases, since this is imposed by the upstream boundary conditions; consequently, the secondary-side steam outlet temperature decreases. In this way, however, the two-phase presence becomes dominant and the heat-transfer coefficient increases. The reason why the exchanged heat settles at a lower value is that the mean temperature difference between the primary and secondary sides decreases much more sharply than the heat-transfer coefficient increases; this happens because the secondary-side profile is tied to its saturation temperature, which keeps the mean temperature nearly unchanged (unlike that of the primary side, which decreases considerably). This phenomenon was not observed in the previous case, in which the pressure constraint no longer held. Clearly, the secondary side, having the same flow rate but being colder, will undergo a smaller enthalpy drop across the turbine, and the power delivered to the grid will be lower. In this scenario, the decrease in the power delivered downstream is caused by the chain of events that begins in the reactor. From this analysis, two different modes of reactor load control can be distinguished, adopted in *PWR*-type reactors (and, in the case under study, in *SMRs*):

1. **Upstream power control.** This consists of a reduction in power, as analyzed above, since the control rods are normally fully withdrawn under nominal operating conditions.

The behavior of the system consistently follows the adjustments made upstream, evolving accordingly and ensuring, downstream, the power level imposed by the control system.

2. **Downstream power control.** This can involve either an increase or a decrease in power (in this treatment, the case of an increase was analyzed) and exploits the *load following* capability characteristic of *PWR*-type reactors. The regulation is carried out by acting on the secondary-circuit turbine, for reasons of logistics and immediacy. The system evolves, however, in such a way as to self-regulate the power generated at the core and make it consistent with that set by the turbine.

The simulation and analysis can also address more realistic and complex situations, in which the two control actions occur simultaneously, and it is essential to predict the behavior of the water temperatures, of the fuel cladding, and of the reactivity, so as to ensure that they remain within the safety limits required for the correct operation of the plant.

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